DEEPCRAFT™ Ready Model for Direction of Arrival (audio)

Introduction

This document describes a DEEPCRAFT™ Ready Model for detecting the direction of arrival of sound, and is developed by Imagimob, an Infineon Technologies company. This model detects the direction from which a given sound is coming from. This report also provides the technical specifications of the machine learning model, its performance in common scenarios, and various test results for the model.

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Model Specification

Model Overview

Upon activation, the Direction of Arrival model can be used, for instance, to locate people talking in a conference room or alert drivers in moving vehicles to a siren in the vicinity. It is important to note that although the model has been trained on specific sounds connected to our audio event detection Ready Models, it tends to recognize the direction of the loudest sound in the environment.

Expected Performance

The model is expected to work in conjunction with a classification model acting as a trigger. Once the sound of interest has been detected, the Direction of Arrival model should be activated to determine its direction. The model can detect eight discrete directions:

- North - N
- Northeast - NE
- East – E
- Southeast – SE
- South – S
- Southwest – SW
- West – W
- Northwest – NW

Each direction corresponds to a 45° angle cone, centered in the direction. The model requires a microphone array of four microphones configured on a + shape (see appendix 1).

The model has an overall accuracy of 87% in real-time test.

Model Tech Specs and Deployment

The DEEPCRAFT™ Ready Model for Direction of Arrival (audio) can detect the direction of a sounds in the following configuration:

- Sample rate: 16000 Hz
- Channels: 4
    - Channel 1: microphone pointing North
    - Channel 2: microphone pointing West
    - Channel 3: microphone pointing South
    - Channel 4: microphone pointing East
- Bit Depth: 32bit

The Flash size of the model is 186 KB and RAM size of the model is 825.5 KB.
The inference time is 82ms on a Cortex-A72.
The model outputs a prediction every 96ms.
The target hardware is Infineon's PSOC™ Edge microcontroller.

Combining the Direction of Arrival with a Classifier

The Direction of Arrival model has no mechanism for differentiating between sounds, and as such it is intended to be used in combination with a classification model or algorithm. The are three options available for utilizing the Direction of Arrival model depending on the user's intention and hardware capability.

Option 1 – Simple Gate
Once the classifier model is activated, enable the Direction of Arrival model for a given duration of time with a clean buffer.

If the trigger event is detected at time t0, the Direction of Arrival model buffer will start to get populated at t0 with new data, which will be used to determine the Direction of Arrival.

This approach is ideal for when you have continuous sounds. This is because the initial data that triggers the classifier is lost to the Direction of Arrival model, which only takes in new data after the classifier is triggered. The benefit here is it is simpler to implement than other methods.

Option 2 – Advanced Gate
Once the trigger model is activated, run the Direction of Arrival model with the same buffer that triggered the classification model.

If the trigger event is detected at time t0, a buffer containing the triggering event will be passed to the Direction of Arrival model.

The buffer size depends on the relative dimensioning of the two models' sliding window.

Let W1 be the size of classification model window and W2 (0.1 seconds) be the size of the Direction of Arrival model window. The value of Δ should be enough to cover the windowing required by the Direction of Arrival model.

- If W1 >= W2 -> Δ = W1
    - Buffer data in the interval [t0 - Δ, t0].
- If W2 > W1, Δ = W2 – W1
    - Buffer data in the interval [t0 - (Δ/2+W1), t0 + Δ/2]

This means that you need to buffer your data and when you have a valid classification you feed the same data that caused the positive classification into the Direction of Arrival model. If the classifier window is smaller, then you need to feed extra data. If it's bigger than W2, feed all the data and look at all the predictions, then you can average the predictions.

This approach is more advanced, but allows you to ensure that the relevant data is processed and that you predict the direction of arrival on exactly the data that triggered the classifier.

Option 3 – Always On
The classification model and the Direction of Arrival model are active at the same time. The audio data streams to both models, but the Direction of Arrival model output is discarded/ignored. Only when the trigger model is activated, is the Direction of Arrival model output enabled for a given duration (or as long as the classification model is triggered).

This approach is less efficient since you always run the Direction of Arrival model, but you only act on the output when you have a valid prediction.

Data Properties

The DEEPCRAFT™ Ready Model for Direction of Arrival (audio) has been constructed using a variety of positive and negative sounds. The positive sounds consisting of different sound of interest, like human voice, coughing, sirens, and alarms. Several instances of the positive data include background noise to enhance model robustness. The negative data corresponds to silence and low background noise. Both positive and negative data have been collected at different distances and directions.

Positive Data

The positive data used to build the model consists of recordings of one event of relevant sounds. These sounds include:

- Alarms
- Car horns
- Gunshots
- Sirens
- Speech
- Drilling

Included Negative Data

The negative data include recordings of silence and low background noise from an office environment.

Testing

Test Results

Validation

The model performance on validation set shows that the model is capable of learning the decision boundary between classes, with an F1-score of above 97%. It's clear that the performance is not homogeneous across the different directions, with a 5% difference between North and South.

Predicted \ Actual	(unlabelled)	E	N	NE	NW	S	SE	SW	W
(unlabelled)	99.91 %	1.92 %	0.48 %	1.73 %	2.24 %	3.98 %	3.27 %	3.13 %	1.65 %
E	0.00 %	97.74 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.93 %
N	0.00 %	0.00 %	99.51 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %
NE	0.03 %	0.00 %	0.00 %	97.76 %	0.00 %	0.00 %	0.00 %	0.13 %	0.00 %
NW	0.00 %	0.00 %	0.00 %	0.00 %	97.45 %	0.02 %	0.20 %	0.00 %	0.00 %
S	0.00 %	0.00 %	0.01 %	0.00 %	0.00 %	94.23 %	0.00 %	0.00 %	0.00 %
SE	0.07 %	0.24 %	0.00 %	0.00 %	0.31 %	0.00 %	96.50 %	0.00 %	0.00 %
SW	0.00 %	0.00 %	0.00 %	0.15 %	0.00 %	0.03 %	0.02 %	96.73 %	0.00 %
W	0.00 %	0.10 %	0.00 %	0.35 %	0.00 %	1.74 %	0.00 %	0.00 %	97.41 %
Total	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %

Display Unit: Normalize

(ACC) Accuracy	97.413 %
(F1S) F1 Score	97.493 %

Class	(unlabelled)	E	N	NE	NW	S	SE	SW	W
(TP) Correct Positive Prediction	14842.000	16999.000	15949.000	17108.000	17193.000	16417.000	16569.000	17302.000	16378.000
(FN) Incorrect Negative Prediction	14.000	393.000	78.000	392.000	449.000	1005.000	601.000	584.000	435.000
(FP) Incorrect Positive Prediction	3203.000	157.000	0.000	28.000	38.000	1.000	105.000	36.000	383.000
(TN) Correct Negative Prediction	134649.000	135159.000	136681.000	135180.000	135028.000	135285.000	135433.000	134786.000	135512.000
(TPR) Recall	99.91 %	97.74 %	99.51 %	97.76 %	97.45 %	94.23 %	96.50 %	96.73 %	97.41 %
(TNR) Selectivity	97.68 %	99.88 %	100.00 %	99.98 %	99.97 %	100.00 %	99.92 %	99.97 %	99.72 %
(PPV) Precision	82.25 %	99.08 %	100.00 %	99.84 %	99.78 %	99.99 %	99.37 %	99.79 %	97.71 %
(NPV) Negative Predictive Value	99.99 %	99.71 %	99.94 %	99.71 %	99.67 %	99.26 %	99.56 %	99.57 %	99.68 %
(FNR) Miss Rate	0.09 %	2.26 %	0.49 %	2.24 %	2.55 %	5.77 %	3.50 %	3.27 %	2.59 %
(FPR) Fall-Out	2.32 %	0.12 %	0.00 %	0.02 %	0.03 %	0.00 %	0.08 %	0.03 %	0.28 %
(FDR) False Discovery Rate	17.75 %	0.92 %	0.00 %	0.16 %	0.22 %	0.01 %	0.63 %	0.21 %	2.29 %

Figure 1: Validation performance breakdown

Real time test

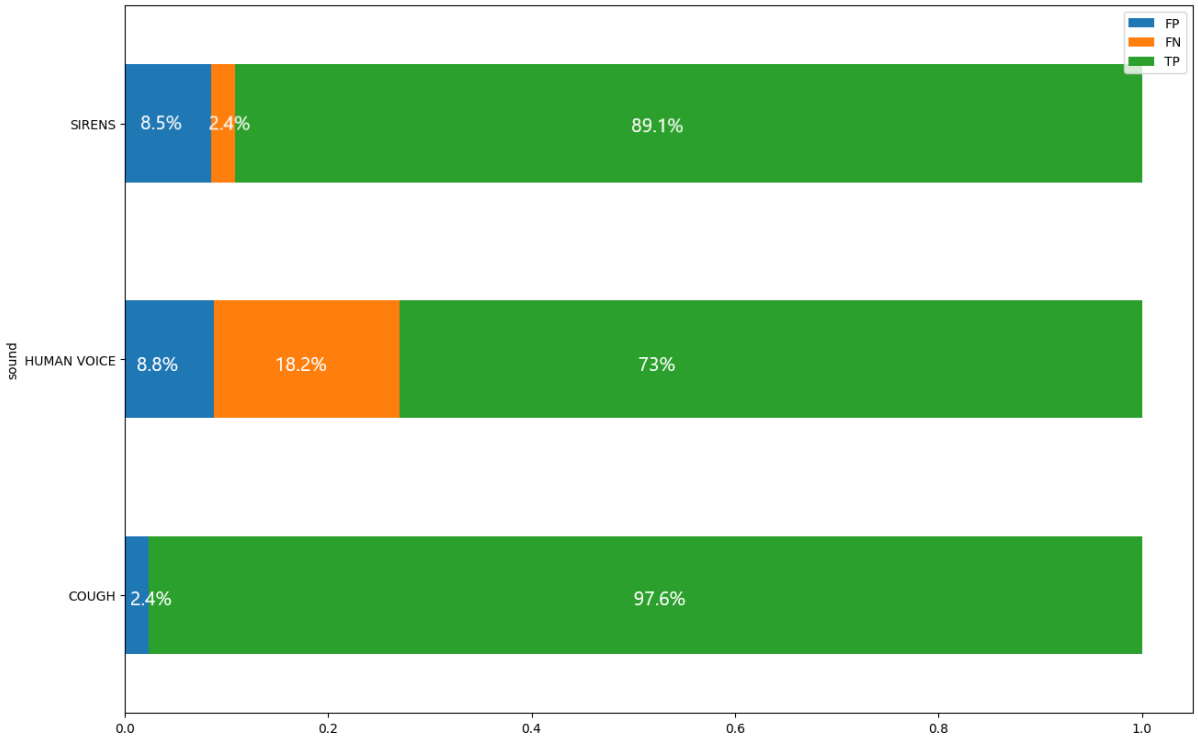
The model has been tested in real-time with both real sources of sounds and sounds played via a speaker. The microphone is placed on a table in the center of the room, then the sound source (person or speaker) is placed at a given distance from the microphone array. The sound source is never below the microphone height. Sounds played via speaker are broken down in segments of fixed duration. A prediction on the segment is considered correct (TP) if the correct direction is detected at least once for the entire duration and no other direction is predicted. The test includes 1487 samples, encompassing various categories of sounds and distances.

- TP = Correct direction predicted
- FP = Wrong direction predicted
- FN = No prediction

Performance Matrix	Actual sound playing
Correct direction predicted	87,2% (1297/1487)
Wrong direction predicted	7,5 % (112/1487)
No prediction	5,3% (78/1487)

True Negatives (TN) are not relevant in this case because the model is expected to work only when a sound is detected by another model, therefore making it meaningless to measure performance in case of silence. This means that in this case, the True Positive Rate (TPR) is the same as the accuracy, which is around 87%.

The bar plot below provides a breakdown of model performance per category. The blue bar and the values on the left side refer to the percentage of samples in each category that trigger a wrong prediction (FP). Sirens and human voice are more subject to false positives. In this testing scenario, one specific sample of siren (appendix II) is responsible for ~79% of the false positives. With respect to human voice, false positives were mostly triggered from noise interleaving moment of silence during speech. The orange bar refers to the percentage of samples that trigger no prediction, which never occurred in case of coughing. However, this is much more frequent with human voice, especially at long distances (see highlighted numbers in table below). This is most likely connected to the gain preprocessing applied to the audio and can be fine-tuned for voice.



SOUND	DISTANCE [M]	FP	FN	TP	TOTAL
COUGH	2	1	0	127	128
	5	5	0	119	124
HUMAN VOICE	1	2	4	70	76
	2	9	6	61	76
	3	7	11	18	36
	5	9	35	75	119
SIRENS	2	39	0	425	464
	5	40	22	402	464

Table 1: Performance by category and distance

Appendix I - Data Sources

Below is the list of data sources outside of Imagimob's own data collection:

- Freesound - https://freesound.org/
- Shapingwaves - https://www.shapingwaves.com/
- Surroundsoundlab - https://www.surroundsoundlab.net/

Appendix II - Factory Sound Examples

A fire truck siren that triggers false positives is provided below:
https://freesound.org/people/Sandermotions/sounds/377126/

Appendix III - Results of Model Testing

Detailed Test Results

SOUNDS:

- C = COUGH
- S = SIRENS
- H = HUMAN VOICE

LOCATIONS:

- L1 = LIVING ROOM (LARGE)
- L2 = OFFICE MEETING ROOM

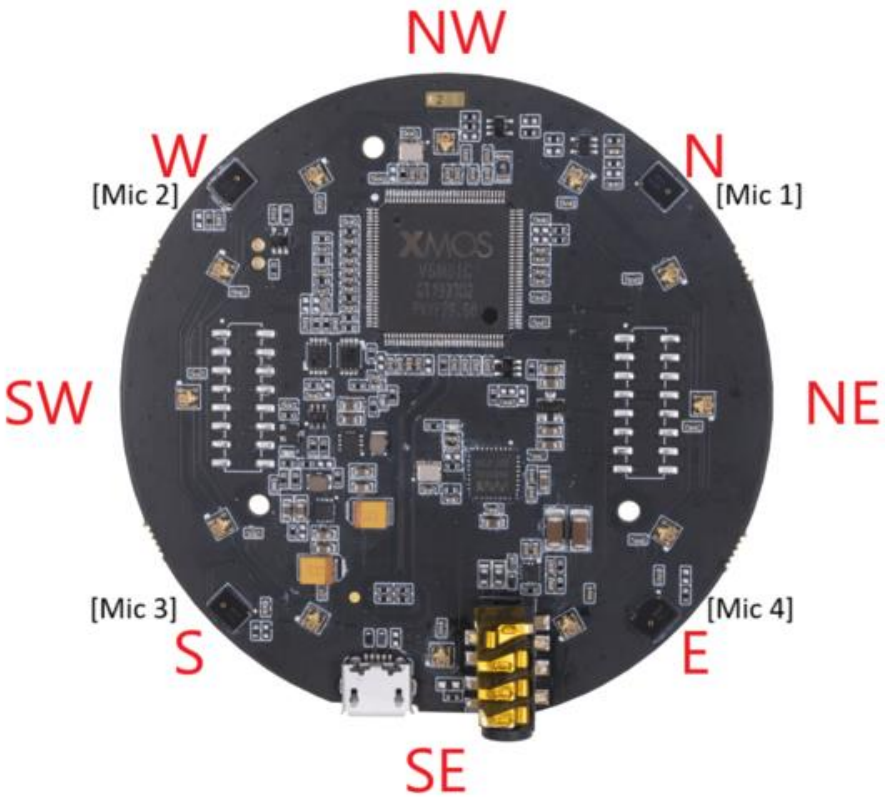
description	sound	noise	distance	FP	FN	TP	TOT.	LOCATION
cough recording #1	C	N	2	0	0	16	16	L1
cough recording #2	C	N	2	1	0	39	40	L1
cough recording #3	C	N	2	0	0	16	16	L1
cough recording #4	C	N	2	0	0	32	32	L1
person coughing #1	C	N	2	0	0	24	24	L1
person coughing #2	C	N	5	0	0	20	20	L1
person coughing #1	C	N	5	0	0	24	24	L1
person coughing #3	C	N	5	1	0	23	24	L1
cough recording #3	C	N	5	1	0	15	16	L1
cough recording #2	C	N	5	3	0	37	40	L1
238228_ambulence-warning-loud	S	N	2	2	0	110	112	L1

377126_fire-truck-small	S	N	2	30	0	26	56	L1
520952_rainy-toronto-with-siren	S	N	2	0	0	72	72	L1
670115_police-siren-echoing-through-c	S	N	2	1	0	63	64	L1
703760_forest-city-edge-air-tone-mixe	S	N	2	6	0	154	160	L1
238228_ambulence-warning-loud	S	N	5	0	14	98	112	L1
377126_fire-truck-small	S	N	5	31	4	21	56	L1
520952_rainy-toronto-with-siren	S	N	5	2	0	70	72	L1
670115_police-siren-echoing-through-c	S	N	5	0	1	63	64	L1
703760_forest-city-edge-air-tone-mixe	S	N	5	7	3	150	160	L1
person talking #1	H	random traffic from window	1	0	0	8	8	L2
person talking #2	H	random traffic from window	1	0	0	8	8	L2
person talking #3	H	random traffic from window	1	0	0	8	8	L2
person talking #4	H	random traffic from window	1	2	0	6	8	L2
person talking #5	H	cafe noise	1	0	1	3	4	L2
person talking #1	H	random traffic from window	2	2	2	4	8	L2
person talking #2	H	random traffic from window	2	0	2	6	8	L2
person talking #3	H	random traffic from window	2	2	0	6	8	L2
person talking #6	H	random traffic from window	2	3	0	5	8	L2
person talking #5	H	random traffic from window	2	0	1	3	4	L2
person talking #1	H	random traffic	3	0	5	3	8	L2

		from window						
person talking #2	H	random traffic from window	3	3	3	2	8	L2
person talking #3	H	random traffic from window	3	1	1	6	8	L2
person talking #6	H	random traffic from window	3	3	1	4	8	L2
person talking #5	H	cafe noise	3	0	1	3	4	L2
person talking #1	H	office environment background noise	1	0	0	8	8	L2
person talking #7	H	office environment background noise	1	0	1	7	8	L2
person talking #8	H	office environment background noise	1	0	0	8	8	L2
person talking #6	H	office environment background noise	1	0	1	7	8	L2
person talking #4	H	office environment background noise	1	0	1	7	8	L2
person talking #1	H	office environment background noise	2	0	0	8	8	L2
person talking #7	H	office environment background noise	2	0	0	8	8	L2
person talking #8	H	office environment backgr	2	0	0	8	8	L2

		ound noise						
person talking #6	H	office environment background noise	2	1	0	7	8	L2
person talking #4	H	office environment background noise	2	1	1	6	8	L2
person talking #9	H	office environment background noise	5	0	2	22	24	L1
person talking #10	H	office environment background noise	5	0	6	17	23	L1
person talking #11	H	office environment background noise	5	3	4	17	24	L1
person talking recording #1	H	office environment background noise	5	3	12	9	24	L1
person talking recording #2	H	office environment background noise	5	3	11	10	24	L1

Appendix IV – Microphone array configuration



Appendix V – Code example

main.c

```c
#include <stdio.h>
#include "audio_data.h"
#include "library.h"
#include "model.h"
#include "time.h"
int data_out[IMAI_DATAOUT_COUNT] = {0};
static const char *LABELS[IMAI_DATAOUT_COUNT] = IMAI_SYMBOL_MAP;

int main(void)
{
    IMAI_DOA_init();

    int num_rows = sizeof(audio_data) / sizeof(audio_data[0]);
    int row_size = sizeof(audio_data[0]) / sizeof(audio_data[0][0]);
    clock_t start;
    clock_t end;
    float duration;
    for (int i = 0; i < num_rows; i++)
    {
        float data_in[4];
        for (int j = 0; j < row_size; j++)
        {
            data_in[j] = (float)audio_data[i][j];
        }
        IMAI_DOA_enqueue(data_in);
        start = clock();
        switch (IMAI_DOA_dequeue(data_out))
        {
        case IMAI_RET_SUCCESS:
            end = clock();
            duration = (float)(end - start) / CLOCKS_PER_SEC;
            for (int i = 1; i < IMAI_DATAOUT_COUNT; i++)
            {
                if (data_out[i] == 1)
                {
                    // printf("%s \r\n", LABELS[i]);
                    printf("Inference time %f\n", duration);
                }
            }
            break;
        }
    }

    return 0;
}
```

audio_data.h

```
const float audio_data[16][4] = {
    { -0.000345, 0.000172, -0.000517, -0.000129 },
    { -0.000560, 0.000216, -0.000474, -0.000172 },
    { -0.000862, 0.000172, -0.000259, -0.000172 },
    { -0.000517, 0.000086, -0.000172, -0.000043 },
    { -0.000388, 0.000172, 0.000000, -0.000216 },
    { -0.000345, 0.000000, 0.000000, -0.000129 },
    { -0.000474, 0.000043, 0.000129, -0.000259 },
    { -0.000259, 0.000302, 0.000302, -0.000086 },
    { -0.000043, 0.000086, 0.000172, -0.000216 },
    { -0.000172, 0.000216, 0.000302, -0.000129 },
    { -0.000000, 0.000086, 0.000388, -0.000129 },
    { -0.000129, -0.000043, 0.000345, -0.000431 },
    { -0.000259, 0.000129, 0.000302, -0.000302 },
    { -0.000043, -0.000259, 0.000043, -0.000216 },
    { 0.000086, -0.000129, 0.000172, -0.000086 },
    { 0.000086, 0.000086, 0.000216, -0.000172 },
};
```